

Vibrations Workshop 9 - Multiple Degrees of Freedom

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ME301

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Problem 1 - Resonate Forcing

The system being analyzed involves two masses equipped with springs and dampers in the configuration below with $m_1 = 10$ kg, $m_2 = 15$ kg, $k_1 = 20$ N/m, $k_2 = 24$ N/m, $k_3 = 22$ N/m, $R = 0.1$ kg/s and two forcing functions f_1, f_2 applied to each of the masses respectively:

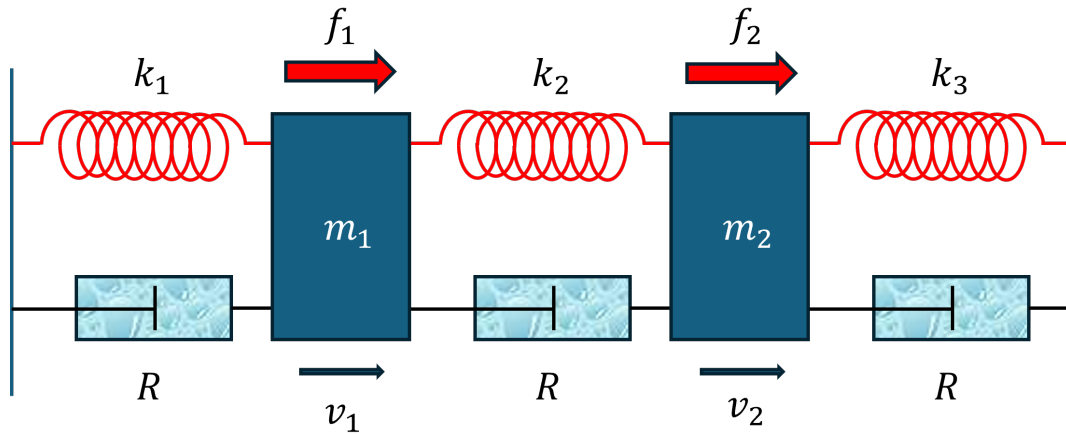


Figure 1: Physical Schematic 1

The above schematic has an equivalent circuit representation in accordance with Kirchhoff's Voltage Law:

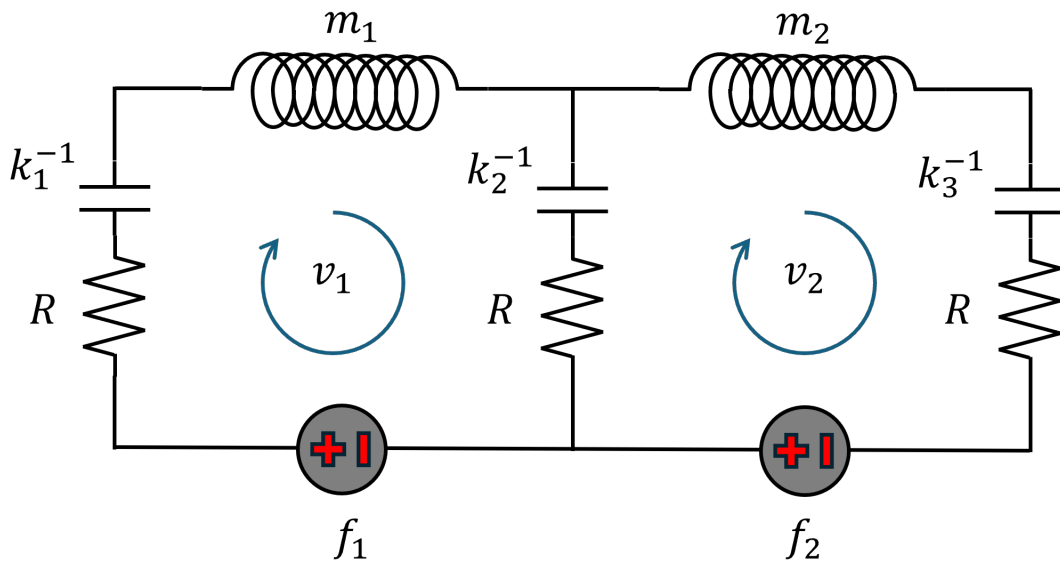


Figure 2: Equivalent Circuit 1

In this regard, the governing equations of motion (derived from Newton's Second Law) can be represented as such in the laplace domain (with $V_i = \mathcal{L}\{v_i\}$ and $\mathcal{L}\{f_i\} = F_i$):

$$\begin{bmatrix} m_1 s + k_1/s + R + k_2/s + R & -R - k_2/s \\ -R - k_2/s & m_2 s + k_2/s + R + k_3/s + R \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$$

The admittance and FRF matrices can thus be identified as:

$$Y(s) = \begin{bmatrix} m_1 s + k_1/s + R + k_2/s + R & -R - k_2/s \\ -R - k_2/s & m_2 s + k_2/s + R + k_3/s + R \end{bmatrix}^{-1} \quad \& \quad H(s) = Y(s)/s$$

Plugging in $s = i\omega$ produces the respective functions that many engineers are very accustomed to. The resultant admittance plots are given below using MATLAB:

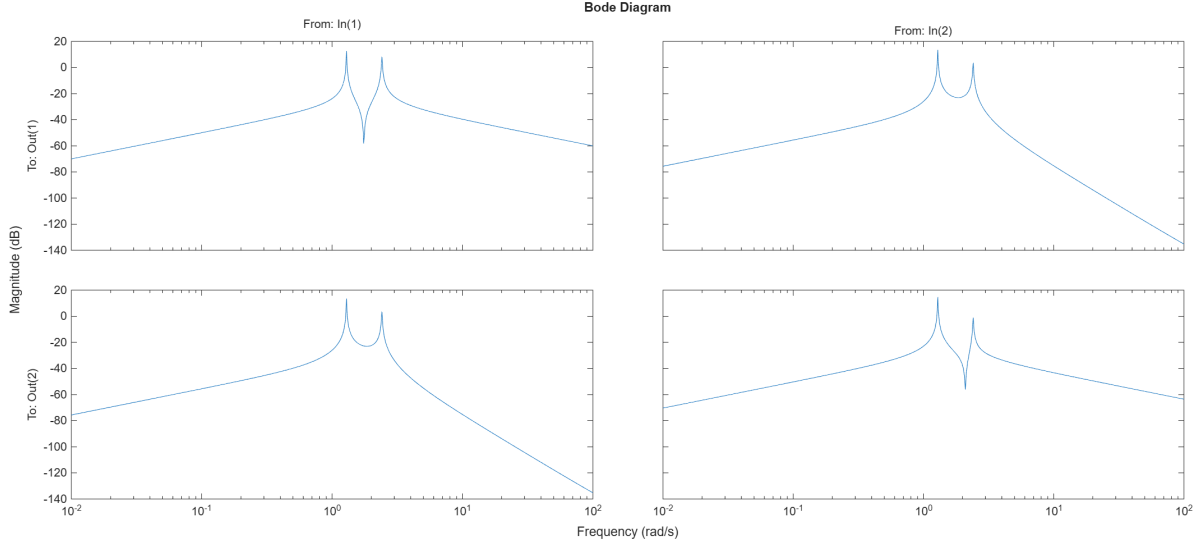


Figure 3: Admittance Magnitude Plots

On the other hand, the frequency response functions (FRFs) likewise have the following plots thanks to MATLAB once again:

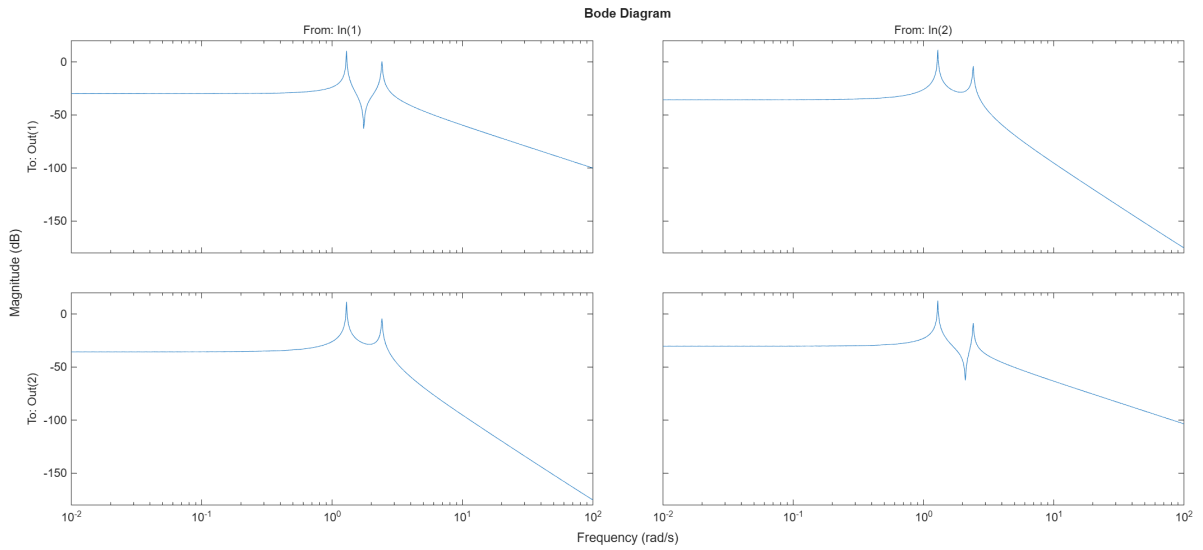


Figure 4: FRF Magnitude Plots

Upon inspection, the resonate frequencies of the system (which corresponds to FRF peaks) occur at $\omega_1 = 1.29$ rad/s and $\omega_2 = 2.41$ rad/s. Likewise, the anti resonate frequencies (which correspond to the FRF valleys) occur at $\omega_{a1} = 1.73$ rad/s and $\omega_{a2} = 2.1$ rad/s.

With all of that out of the way, suppose that the system is forced externally by the forcing functions of the following form:

$$f_i(t) = (1 \text{ N}) \sin(\omega t),$$

where ω is some frequency to be determined later. Note that if $\omega = 0$, $f_i(t) = 0$.

In order to properly incorporate these time dependent forces into the analysis, they will be brought into the laplace domain. In particular:

$$\begin{aligned} F_i(s) &= \mathcal{L}\{f_i\}(s) \\ &= \int_0^\infty e^{-st} f_i(t) dt = \int_0^\infty e^{-st} (1 \text{ N}) \sin(\omega t) dt = (1 \text{ N}) \int_0^\infty e^{-st} \sin(\omega t) dt \\ &= (1 \text{ N}) \int_0^\infty e^{-st} \frac{e^{i\omega t} - e^{-i\omega t}}{2i} dt = \frac{1 \text{ N}}{2i} \int_0^\infty e^{(-s+i\omega)t} + e^{(-s-i\omega)t} dt \\ &= \frac{1 \text{ N}}{2i} \left(\frac{e^{(-s+i\omega)t}}{-s+i\omega} \Big|_{t=0}^\infty + \frac{e^{(-s-i\omega)t}}{-s-i\omega} \Big|_{t=0}^\infty \right) = \frac{1 \text{ N}}{2i} \left(\frac{1}{s-i\omega} + \frac{1}{s+i\omega} \right) \\ &= \frac{(1 \text{ N})\omega}{s^2 + \omega^2} \quad \text{for} \quad \text{Re}(s) > 0 \end{aligned}$$

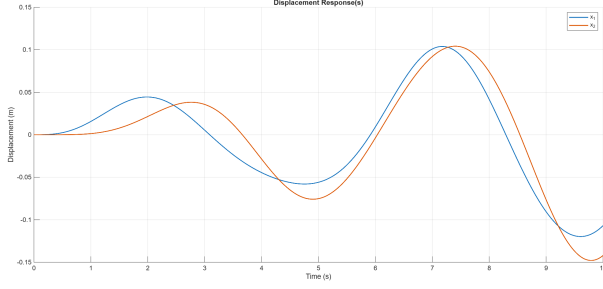
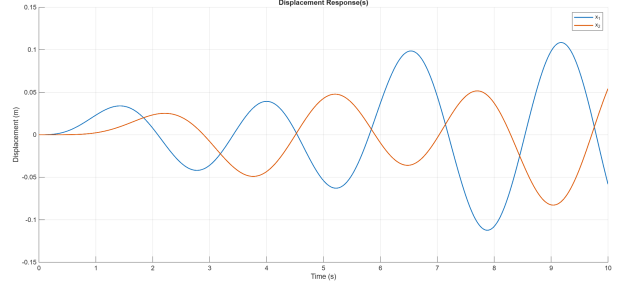
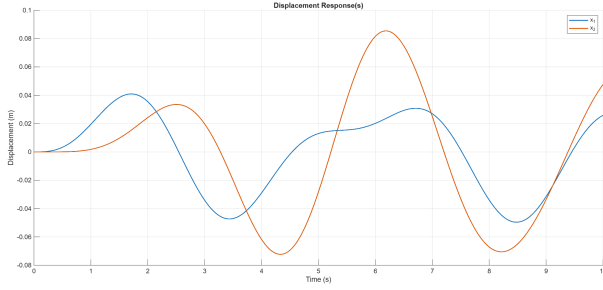
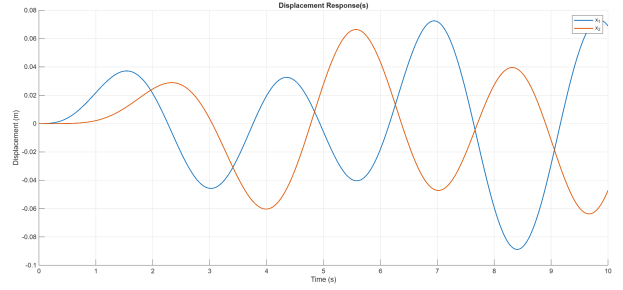
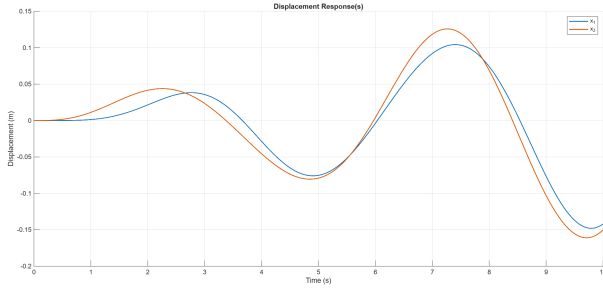
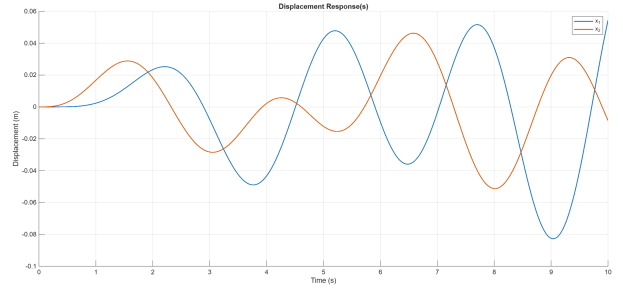
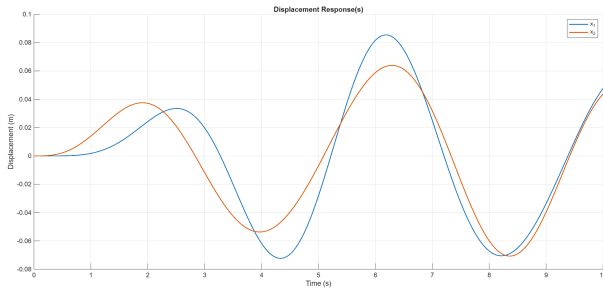
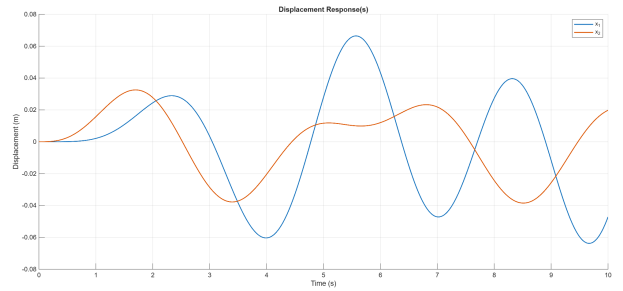
Using MATLAB, the corresponding trajectories x_1, x_2 can be plotted against time under the assumption that $x_i(0) = 0 = \dot{x}_i(0)$. Note that $(x_1, x_2)(\alpha, \beta)$ corresponds to the x_1, x_2 responses to $f_1(t) = (1 \text{ N}) \sin(\alpha t)$ and $f_2(t) = (1 \text{ N}) \sin(\beta t)$.

To be specific, the way in which the trajectories are determined (analytically) is as follows:

$$x_j(t) = \mathcal{L}_s^{-1} \left\{ \sum_{k=1}^2 Y_{jk}(s) F_k(s) / s \right\} (t) = \frac{1}{2\pi i} \sum_{k=1}^2 \int_{\gamma-i\infty}^{\gamma+i\infty} \frac{Y_{jk}(s) F_k(s)}{s} e^{st} ds$$

In reality though, MATLAB (lsim) and other ODE solver methods uses numerical integration methods to work out the inverse laplace transforms, which is less cool, but more efficient.

Anyway, the trajectories x_1, x_2 are plotted below using the labeling convention above for reference:

Figure 5: Displacement Response: $(x_1, x_2)(\omega_1, 0)$ Figure 6: Displacement Response: $(x_1, x_2)(\omega_2, 0)$ Figure 7: Displacement Response: $(x_1, x_2)(\omega_{a1}, 0)$ Figure 8: Displacement Response: $(x_1, x_2)(\omega_{a2}, 0)$ Figure 9: Displacement Response: $(x_1, x_2)(0, \omega_1)$ Figure 10: Displacement Response: $(x_1, x_2)(0, \omega_2)$ Figure 11: Displacement Response: $(x_1, x_2)(0, \omega_{a1})$ Figure 12: Displacement Response: $(x_1, x_2)(0, \omega_{a2})$

Interestingly, when either mass is forced with frequency ω_1 , both masses are more or less in phase. When either mass is forced with frequency ω_2 , both masses are out of phase. When either mass is forced with frequency ω_{a1} or ω_{a2} , both masses destructively interfere with their trajectories resulting in low amplitude time intervals.

More specifically, the motion of the two masses changes substantially depending on whether the forcing

frequency corresponds to a resonant or anti-resonant condition. In particular, the relative phase between the masses and the amplitude distribution throughout the system vary significantly between cases.

Case i: $f_1(t) = \sin(\omega_1 t)$, $f_2(t) = 0$

At the first resonant frequency ω_1 , both masses oscillate approximately in phase. The displacement plots show that both masses move in the same direction simultaneously for most of the motion. Physically, this corresponds to the first mode shape of the coupled system, where the spring coupling between the masses is not sufficiently stiff to force opposing motion.

Because the excitation is applied directly to mass m_1 , the response amplitude of m_1 is slightly larger initially; however, energy transfers through the coupling spring k_2 , causing m_2 to quickly participate in the motion. The amplitudes of both masses become comparatively large due to resonance.

Case ii: $f_1(t) = \sin(\omega_2 t)$, $f_2(t) = 0$

At the second resonant frequency ω_2 , the masses oscillate predominantly out of phase. When m_1 moves upward, m_2 moves downward, and vice versa. This behavior is characteristic of the second mode shape of the system.

The coupling spring k_2 experiences substantially larger deformation in this mode because the relative displacement between the masses is much greater. Consequently, the internal restoring forces become stronger than in the first mode.

Although resonance still produces large amplitudes, the opposing motion causes the center of mass of the system to move less than in the first resonant case.

Case iii: $f_1(t) = \sin(\omega_{a1} t)$, $f_2(t) = 0$

At the first anti-resonant frequency ω_{a1} , the displacement response becomes strongly suppressed despite continuous forcing. The motion of the masses interferes destructively, reducing the overall displacement amplitude.

In this regime, the force transmitted through the coupling spring and dampers produces partial cancellation of the external excitation. As a result, one mass may continue oscillating while the other exhibits comparatively little motion.

Case iv: $f_1(t) = \sin(\omega_{a2} t)$, $f_2(t) = 0$

At the second anti-resonant frequency ω_{a2} , destructive interference again occurs, though the distribution of amplitudes differs from the previous anti-resonant case.

The relative phase between the masses shifts in such a way that energy introduced into the system is re-distributed internally rather than producing large observable displacement amplitudes. Compared to the resonant cases, the trajectories remain significantly bounded.

Case v: $f_1(t) = 0$, $f_2(t) = \sin(\omega_1 t)$

When the forcing is moved to mass m_2 while the system is excited at the first resonant frequency, the qualitative behavior remains similar to Case i. Both masses still oscillate largely in phase because the excitation frequency aligns with the first mode shape.

However, the amplitude distribution shifts toward m_2 , since the external energy is now injected directly into that coordinate. Energy is then transferred through the coupling spring to m_1 .

Case vi: $f_1(t) = 0$, $f_2(t) = \sin(\omega_2 t)$

At the second resonant frequency with forcing applied to m_2 , the system again exhibits predominantly out-of-phase motion.

The deformation of the coupling spring becomes large because the masses move in opposite directions. The displacement amplitudes remain large due to resonance, but the motion is dominated by internal deformation rather than collective translation.

Case vii: $f_1(t) = 0$, $f_2(t) = \sin(\omega_{a_1}t)$

At the first anti-resonant frequency with forcing applied to m_2 , destructive interference again suppresses the response amplitude.

The excitation energy is partially canceled through the coupled dynamics of the system. The response therefore contains intervals where the displacement amplitudes become very small even though forcing continues.

Case viii: $f_1(t) = 0$, $f_2(t) = \sin(\omega_{a_2}t)$

At the second anti-resonant frequency, the system once again displays reduced displacement amplitudes due to destructive interference between coupled motions.

Unlike resonance, where energy accumulates constructively, anti-resonance redistributes energy internally in a manner that suppresses motion at particular coordinates. Consequently, the system can exhibit relatively small observable displacements even under sustained harmonic excitation.

Problem 2 - The Unconstrained & Unbalanced Seesaw

The system being analyzed here involves a rigid rod (approximately) of length $l = 2.6$ m and mass $m = 124$ kg (uniformly distributed). At each end of the rod, springs with respective stiffness's of $k_1 = 1000$ N/m and $k_2 = 3500$ N/m. This described physical system is illustrated below with labeled coordinates:

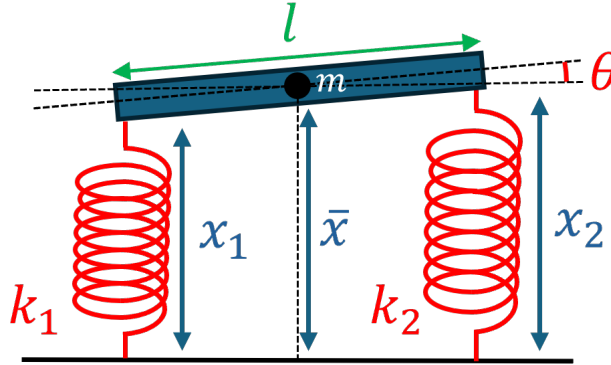


Figure 13: Physical Schematic 2

If not clear the labeled coordinates have the following relationships:

$$\bar{x} = \frac{x_1 + x_2}{2} \quad \text{and} \quad \theta = \sin^{-1} \left(\frac{x_2 - x_1}{l} \right) \approx \frac{x_2 - x_1}{l} \quad \text{for} \quad x_1 \approx x_2$$

The moment of inertia of the rod is given by:

$$I = \int_{-l/2}^{l/2} r^2 \frac{m}{l} dr = \frac{1}{12} ml^2$$

The kinetic energy of the system can be naturally expressed as such:

$$\begin{aligned}
K &= \frac{1}{2}m\dot{x}^2 + \frac{1}{2}I\dot{\theta}^2 = \frac{1}{2}m\dot{x}^2 + \frac{1}{24}ml^2\dot{\theta}^2 \\
&= \frac{1}{2}m\left(\frac{\dot{x}_1 + \dot{x}_2}{2}\right)^2 + \frac{1}{24}ml^2\left(\frac{\dot{x}_2 - \dot{x}_1}{l\sqrt{1 - \left(\frac{x_2 - x_1}{l}\right)^2}}\right)^2 \\
&= \frac{1}{8}m(\dot{x}_1 + \dot{x}_2)^2 + \frac{1}{24}m\frac{(\dot{x}_2 - \dot{x}_1)^2}{1 - \left(\frac{x_2 - x_1}{l}\right)^2} \approx \frac{1}{8}m(\dot{x}_1 + \dot{x}_2)^2 + \frac{1}{24}m(\dot{x}_2 - \dot{x}_1)^2 \quad \text{for } x_1 \approx x_2
\end{aligned}$$

The potential energy of the system (neglecting gravity) can be naturally expressed as such:

$$\begin{aligned}
U &= \frac{1}{2}k_1\left(\bar{x} - \frac{l}{2}\sin\theta\right)^2 + \frac{1}{2}k_2\left(\bar{x} + \frac{l}{2}\sin\theta\right)^2 \\
&= \frac{1}{2}k_1\left(\frac{x_1 + x_2}{2} - \frac{x_2 - x_1}{2}\right)^2 + \frac{1}{2}k_2\left(\frac{x_1 + x_2}{2} + \frac{x_2 - x_1}{2}\right)^2 \\
&= \frac{1}{2}k_1x_1^2 + \frac{1}{2}k_2x_2^2 \\
&\quad (\text{and note}) \approx \frac{1}{2}k_1(\bar{x} - l\theta/2)^2 + \frac{1}{2}k_2(\bar{x} + l\theta/2)^2 \quad \text{for } \theta \approx 0
\end{aligned}$$

Now, using the coordinates x_1, x_2 in conjunction with the small deflection approximation ($x_1 \approx x_2$), the lagrangian of the system can be written out as such:

$$\mathcal{L} = K - U = \frac{1}{8}m(\dot{x}_1 + \dot{x}_2)^2 + \frac{1}{24}m(\dot{x}_2 - \dot{x}_1)^2 - \frac{1}{2}k_1x_1^2 - \frac{1}{2}k_2x_2^2$$

x_1 Momentum & Force:

$$\frac{\partial \mathcal{L}}{\partial \dot{x}_1} = \frac{1}{4}m(\dot{x}_1 + \dot{x}_2) + \frac{1}{12}m(\dot{x}_1 - \dot{x}_2) = \frac{1}{3}m\dot{x}_1 + \frac{1}{6}m\dot{x}_2 \quad \& \quad \frac{\partial \mathcal{L}}{\partial x_1} = -k_1x_1$$

x_2 Momentum & Force:

$$\frac{\partial \mathcal{L}}{\partial \dot{x}_2} = \frac{1}{4}m(\dot{x}_1 + \dot{x}_2) + \frac{1}{12}m(\dot{x}_2 - \dot{x}_1) = \frac{1}{3}m\dot{x}_2 + \frac{1}{6}m\dot{x}_1 \quad \& \quad \frac{\partial \mathcal{L}}{\partial x_2} = -k_2x_2$$

Euler Lagrange Equations:

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}_i} - \frac{\partial \mathcal{L}}{\partial x_i} = 0$$

Equations of Motion:

$$\frac{1}{3}m\ddot{x}_1 + \frac{1}{6}m\ddot{x}_2 + k_1x_1 = 0 \quad \& \quad \frac{1}{6}m\ddot{x}_1 + \frac{1}{3}m\ddot{x}_2 + k_2x_2 = 0$$

Matrix Representation:

$$\begin{bmatrix} \frac{1}{3}m & \frac{1}{6}m \\ \frac{1}{6}m & \frac{1}{3}m \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} + \begin{bmatrix} k_1 & 0 \\ 0 & k_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

Applying the (linear) Laplace Transform:

$$\begin{bmatrix} \frac{1}{3}ms + k_1/s & \frac{1}{6}ms \\ \frac{1}{6}ms & \frac{1}{3}ms + k_2/s \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = (\text{some function of the initial conditions})$$

The Impedance Matrix:

$$Z_{(x_1, x_2)}(s) = \begin{bmatrix} \frac{1}{3}ms + k_1/s & \frac{1}{6}ms \\ \frac{1}{6}ms & \frac{1}{3}ms + k_2/s \end{bmatrix}$$

Using MATLAB, the resultant admittance ($Y = Z^{-1}$) plots can be produced. They are shown below:

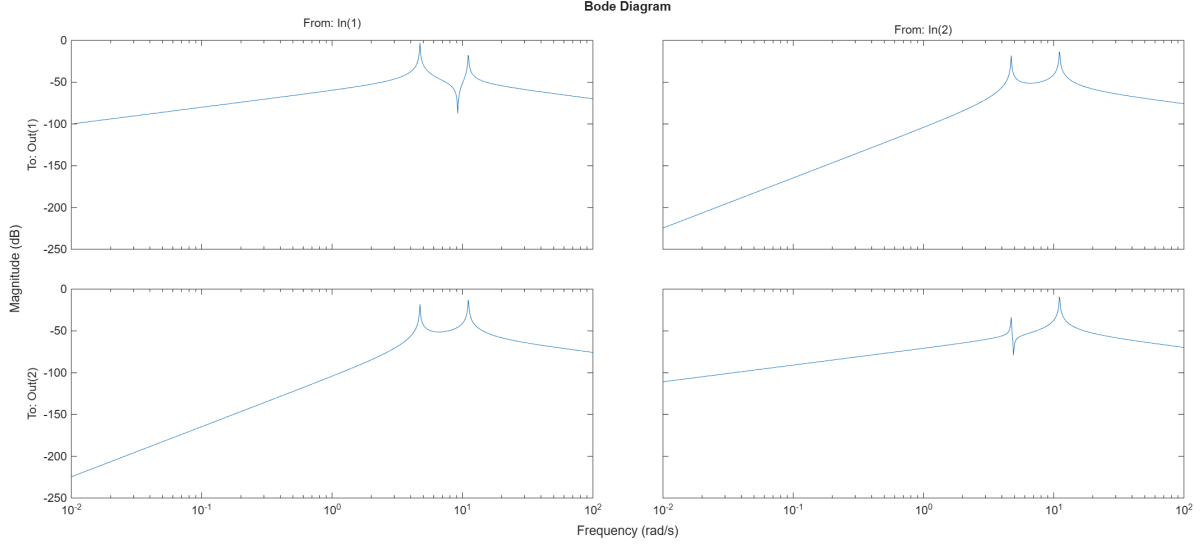


Figure 14: Admittance Magnitude Plots Using (x_1, x_2) Coordinates

Again, using MATLAB, the FRF ($H(s) = Y(s)/s$) can be produced. Its components are likewise plotted below in the frequency domain:

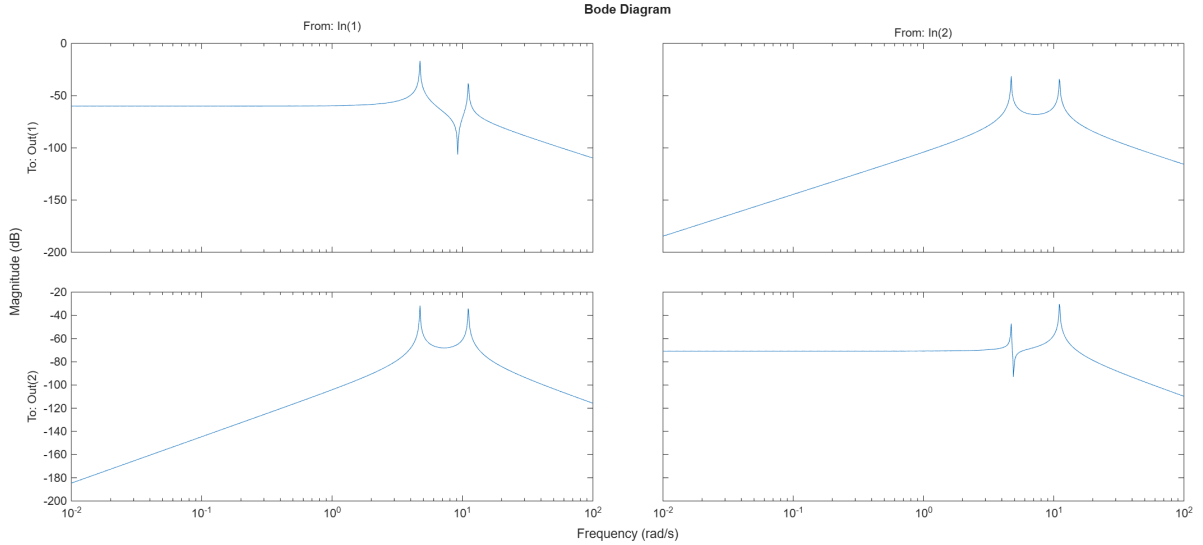


Figure 15: FRF Magnitude Response Plots Using (x_1, x_2) Coordinates

The equations of motion can be determined in the other set of coordinates $(\bar{x}, \theta)$ as well. In particular, using the small deflection approximation ($\theta \approx 0$), the lagrangian of the system can be written out as such:

$$\mathcal{L} = K - U = \frac{1}{2}m\dot{\bar{x}}^2 + \frac{1}{24}ml^2\dot{\theta}^2 - \frac{1}{2}k_1(\bar{x} - l\theta/2)^2 - \frac{1}{2}k_2(\bar{x} + l\theta/2)^2$$

$\bar{x}$ Momentum and Force:

$$\frac{\partial \mathcal{L}}{\partial \dot{\bar{x}}} = m\dot{\bar{x}} \quad \& \quad \frac{\partial \mathcal{L}}{\partial \bar{x}} = -k_1(\bar{x} - l\theta/2) - k_2(\bar{x} + l\theta/2) = -(k_1 + k_2)\bar{x} - \frac{1}{2}(k_2 - k_1)l\theta$$

θ Angular Momentum and Torque:

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = \frac{1}{12}ml^2\dot{\theta} \quad \& \quad \frac{\partial \mathcal{L}}{\partial \theta} = \frac{1}{2}k_1l(\bar{x} - l\theta/2) - \frac{1}{2}k_2l(\bar{x} + l\theta/2) = -\frac{1}{2}(k_2 - k_1)l\bar{x} - \frac{1}{4}(k_1 + k_2)l^2\theta$$

Equations of Motion:

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\bar{x}}} - \frac{\partial \mathcal{L}}{\partial \bar{x}} = 0 \quad \& \quad \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} - \frac{\partial \mathcal{L}}{\partial \theta} = 0$$

(alternatively)

$$m\ddot{\bar{x}} + (k_1 + k_2)\bar{x} + \frac{1}{2}(k_2 - k_1)l\theta = 0 \quad \& \quad \frac{1}{12}ml^2\ddot{\theta} + \frac{1}{2}(k_2 - k_1)l\bar{x} + \frac{1}{4}(k_1 + k_2)l^2\theta = 0$$

Matrix Representation:

$$\begin{bmatrix} m & 0 \\ 0 & \frac{1}{12}ml^2 \end{bmatrix} \begin{bmatrix} \ddot{\bar{x}} \\ \ddot{\theta} \end{bmatrix} + \begin{bmatrix} k_1 + k_2 & \frac{1}{2}(k_2 - k_1)l \\ \frac{1}{2}(k_2 - k_1)l & \frac{1}{4}(k_1 + k_2)l^2 \end{bmatrix} \begin{bmatrix} \bar{x} \\ \theta \end{bmatrix} = 0$$

Applying the (linear) Laplace Transform:

$$\begin{bmatrix} ms + (k_1 + k_2)/s & \frac{1}{2}(k_2 - k_1)l/s \\ \frac{1}{2}(k_2 - k_1)l/s & \frac{1}{12}ml^2s + \frac{1}{4}(k_1 + k_2)l^2 \end{bmatrix} \begin{bmatrix} \bar{V} \\ \Omega \end{bmatrix} = (\text{some function of the initial conditions})$$

The Impedance Matrix:

$$Z_{(\bar{x}, \theta)}(s) = \begin{bmatrix} ms + (k_1 + k_2)/s & \frac{1}{2}(k_2 - k_1)l/s \\ \frac{1}{2}(k_2 - k_1)l/s & \frac{1}{12}ml^2s + \frac{1}{4}(k_1 + k_2)l^2 \end{bmatrix}$$

Using MATLAB, the resultant admittance ($Y = Z^{-1}$) plots can be produced. They are shown below:

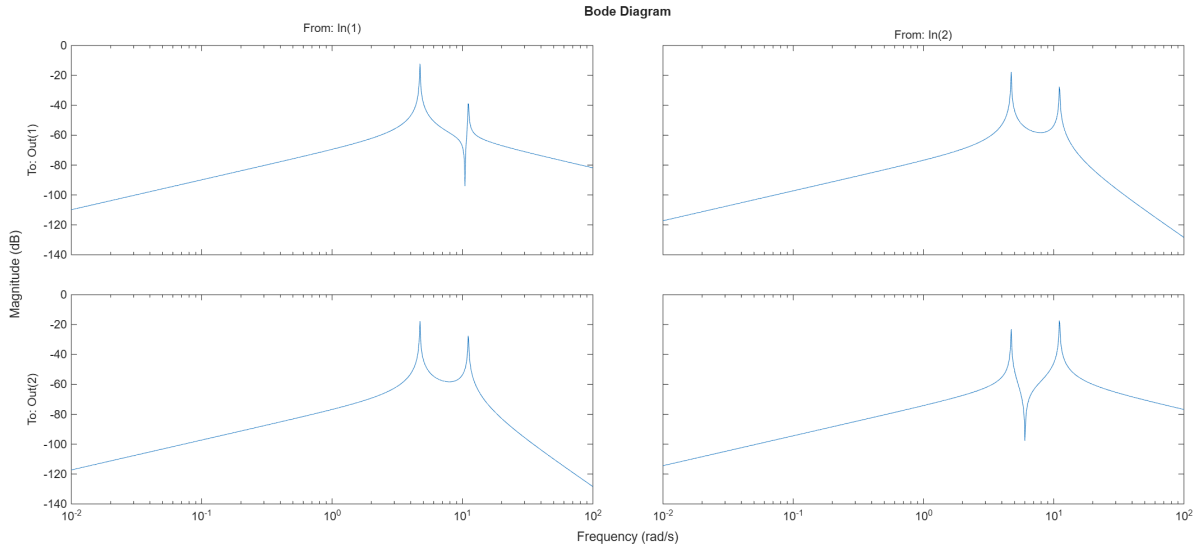


Figure 16: Admittance Magnitude Plots Using $(\bar{x}, \theta)$ Coordinates

Again, using MATLAB, the FRF ($H(s) = Y(s)/s$) can be produced. Its components are likewise plotted below in the frequency domain:

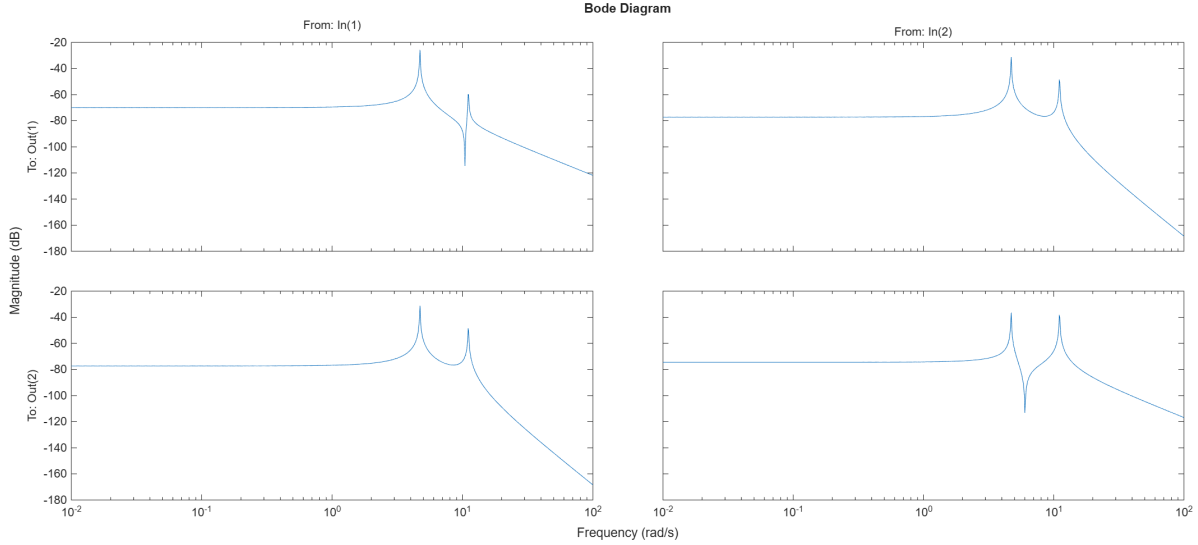


Figure 17: FRF Magnitude Plots Using $(\bar{x}, \theta)$ Coordinates

Amazingly, between both coordinate systems, their (collective) respective FRF plots have the exact same resonate frequencies! This is due to the fact that the underlying physics is coordinate invariant.

More specifically, the resonance frequencies obtained using the (x_1, x_2) coordinates and the $(\bar{x}, \theta)$ coordinates are identical because both coordinate systems describe the same physical system. Changing coordinates does not alter the underlying eigenvalues of the coupled equations of motion, and therefore the natural frequencies remain invariant.

However, although the resonant frequencies are identical, the anti-resonant frequencies are not necessarily identical between the two formulations. This occurs because anti-resonance depends not only on the intrinsic dynamics of the system, but also on the particular input-output relationship being observed.

In the (x_1, x_2) coordinate system, the FRFs correspond directly to translational motion at the two ends of the rod. The transfer functions therefore measure how forcing applied to one endpoint produces motion at another endpoint. Destructive interference between translational coordinates produces anti-resonant frequencies at particular values of ω .

In contrast, the $(\bar{x}, \theta)$ coordinate system separates the motion into center-of-mass translation and rotational motion. The FRFs in this representation therefore correspond to fundamentally different physical observables. Since anti-resonance depends on cancellation within a specific transfer function, changing the observed coordinates changes the locations of these cancellations.

Consequently, the resonant frequencies remain invariant because they are determined by the poles of the system, whereas the anti-resonant frequencies change because they are determined by the zeros of the specific transfer functions being measured.

More concretely, the Euler Lagrange Equations in any coordinate system are equivalent to the following coordinate invariant condition:

Appendix: MATLAB Code

```

1  clc; clear; close all;
2
3  %% Parameters
4

```

```

5  % Part 1
6  m1 = 10;
7  m2 = 15;
8  k1 = 20;
9  k2 = 24;
10 k3 = 22;
11 R = 0.1;
12 omega = [1.29  2.41  1.73  2.1];
13 omega = omega(4);
14
15 % Part 2
16 m = 124;
17 l = 2.6;
18 k1 = 1000;
19 k2 = 3500;
20
21 %% Impedance Matrix
22
23 % Part 1
24 Z = @(s) ...
25     [m1*s+k1/s+R+k2/s+R, -R-k2/s;
26      -R-k2/s, m2*s+k2/s+R+k3/s+R];
27
28 % Part 2a
29 Z = @(s) ...
30     [(m/3)*s+k1/s, (m/6)*s;
31      (m/6)*s, (m/3)*s+k2/s];
32
33 % Part 2b
34 Z = @(s) ...
35     [m*s+(k1+k2)/s, 0.5*(k2-k1)*l/s;
36      0.5*(k2-k1)*l/s, (1/12)*m*(l^2)*s + 0.25*(k1+k2)*l^2/s];
37
38 %% Control System Toolbox Implementation
39 s = tf('s');
40 Z_tf = Z(s);
41
42 %% Laplaced External Force Vector & Initial Conditions
43 F = [0 omega/(s^2+omega^2)];
44 x0 = [0 0];
45 v0 = [0 0];
46
47 %% Time Domain Parameters
48 t0 = 0;
49 t1 = 10;
50 dt = 0.001;
51
52 %% Coefficient Extraction
53 syms z
54 Z_sym = Z(z);
55
56 n = size(Z(1),1);
57 for i = 1:n
58     for j = 1:n

```

```

59     Zij = Z_sym(i,j);
60     M(i,j) = double(limit(Zij/z,z,+inf));
61     K(i,j) = double(limit(Zij*z,z,0));
62     R(i,j) = double(limit(Zij - M(i,j)*z - K(i,j)/z, z, 1));
63 end
64 end
65
66 %% Augmented Force Vector
67 F_aug = F' + M*v0' - (K*x0')/s;
68
69 %% Dependent Laplace Variables
70 Y = inv(Z_tf);
71 V = Y*F_aug;
72 X = V/s + x0'/s;
73 H = Y/s;
74
75 %% Time-Domain (and frequency)
76 t = t0:dt:t1;
77 u = zeros(size(t));
78 u(1) = 1/(t(2)-t(1));
79 w = logspace(-2,2,1000);
80 [v_resp, t_out] = lsim(V, u, t);
81 [x_resp, ~] = lsim(X, u, t);
82
83 % Extract signals
84 v = squeeze(v_resp(:, :, 1));
85 x = squeeze(x_resp(:, :, 1));
86
87 %% Plot Velocities
88 figure; hold on; grid on;
89 for k = 1:n
90     plot(t_out, v(:,k), 'LineWidth',1.4);
91 end
92 xlabel('Time (s)');
93 ylabel('Velocity (m/s)');
94 title('Velocity Response(s)');
95 legend(arrayfun(@(k) sprintf('v_%d',k),1:n,'UniformOutput',false));
96
97 %% Plot Displacements
98 figure; hold on; grid on;
99 for k = 1:n
100     plot(t_out, x(:,k), 'LineWidth',1.4);
101 end
102 xlabel('Time (s)');
103 ylabel('Displacement (m)');
104 title('Displacement Response(s)');
105 legend(arrayfun(@(k) sprintf('x_%d',k),1:n,'UniformOutput',false));
106
107 %%% BODE Plot (Y)
108 %
109 % figure; hold on; grid on;
110 % legend_entries = {};
111 %
112 % for i = 1:n

```

```

113 %      for j = 1:n
114 %
115 %          [mag,phase] = bode(Y(i,j), w);
116 %
117 %          mag    = squeeze(mag);
118 %          phase  = squeeze(phase);
119 %
120 %          subplot(2,1,1); hold on; grid on;
121 %          semilogx(w,20*log10(mag), 'LineWidth',1.4);
122 %          ylabel('Modulus (dB)');
123 %          title('Admittance Bode Plot');
124 %
125 %          subplot(2,1,2); hold on; grid on;
126 %          semilogx(w,phase, 'LineWidth',1.4);
127 %          ylabel('Phase (deg)');
128 %          xlabel('\omega (rad/s)');
129 %
130 %          legend_entries{end+1} = sprintf('Y-{%d%d}',i,j);
131 %      end
132 % end
133 %
134 % subplot(2,1,1); legend(legend_entries, 'Location', 'best');
135 % subplot(2,1,2); legend(legend_entries, 'Location', 'best');
136 %
137 % %% BODE Plot (H)
138 % figure; hold on; grid on;
139 % legend_entries = {};
140 %
141 % for i = 1:n
142 %     for j = 1:n
143 %
144 %         [mag,phase] = bode(H(i,j), w);
145 %
146 %         mag    = squeeze(mag);
147 %         phase  = squeeze(phase);
148 %
149 %         subplot(2,1,1); hold on; grid on;
150 %         semilogx(w,20*log10(mag), 'LineWidth',1.4);
151 %         ylabel('Modulus (dB)');
152 %         title('Frequency Response Bode Plot');
153 %
154 %         subplot(2,1,2); hold on; grid on;
155 %         semilogx(w,phase, 'LineWidth',1.4);
156 %         ylabel('Phase (deg)');
157 %         xlabel('\omega (rad/s)');
158 %
159 %         legend_entries{end+1} = sprintf('H-{%d%d}',i,j);
160 %     end
161 % end
162 %
163 % subplot(2,1,1); legend(legend_entries, 'Location', 'best');
164 % subplot(2,1,2); legend(legend_entries, 'Location', 'best');
165 %
166 % %% Test

```

```
167 figure ;  
168 bodemag(Y,w) ;  
169 figure ;  
170 bodemag(H,w) ;
```